UNIVERSITY OF UYO

FACULTY OF SCIENCE

DEPARTMENT OF MATHEMATICS

ASECOND Semester Examination 2024/2025 Session

Course Title: Metric Space Topology **Course Code:** MTH321

Instructions: **Answer any Three Questions** Time: **2 Hours**

1. (a) Define the following concepts (i) concept of a metric function and explain its properties, (ii) metric space, (iii) Metric space topology.
(b) Define norm function on a metric space and explain its properties
(c) Give two examples each for (a) and (b) above

2.Prove that every normed linear space is a metric space

3. (a) Define the concept of conjugate exponents and explain with examples
(b) (i) State (i) The Young inequality, (ii) the Scwartz and (iii) the Minkowski inequality in R^2.

4. Prove that the function $d_2 = |.|_2$ is a metric function on R^2

5. Write short notes on (a) homeomorphism, (b) covering of a set, (c) open and closed sets in a metric space, (d) open and closed balls, (e) limit points of a set A in a metric space in a metric space

CamScanner